

QCE Chemistry Units 3&4

Paper 1

General instruction

- Work in this booklet will not be marked.

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SECTION 1

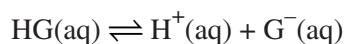
QUESTION 1

Which of the following is an amide?

- (A) $\text{CH}_3\text{CH}_2\text{NH}_2$
- (B) CH_3CONH_2
- (C) $\text{CH}_3\text{OCH}_2\text{NH}_2$
- (D) $\text{CH}_3\text{CH}_2\text{CN}$

QUESTION 2

Bromocresol green (HG) is an indicator and a weak acid with a K_a of 2.0×10^{-5} . It is yellow in its acidic form and green in its basic form. The equilibrium reaction is as follows.



If a few drops of bromocresol green are added to an acidic solution with pH 2.0, the position of the equilibrium in the reaction shown

- (A) shifts right and the solution turns green.
- (B) shifts right and the solution turns yellow.
- (C) shifts left and the solution turns green.
- (D) shifts left and the solution turns yellow.

QUESTION 3

At 40°C , the hydronium ion concentration in pure water is $1.71 \times 10^{-7} \text{ M}$.

Pure water at 40°C will therefore

- (A) be acidic with a pH of 6.8.
- (B) contain hydroxide ions at a concentration of $5.85 \times 10^{-8} \text{ M}$.
- (C) contain fewer ions than water at 25°C .
- (D) have a K_w value of $2.92 \times 10^{-14} \text{ M}^2$.

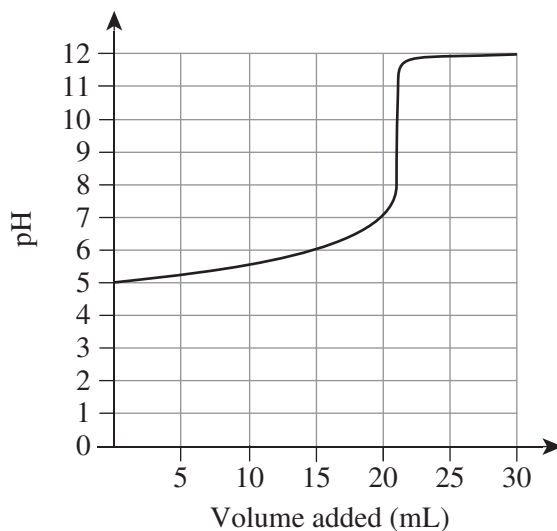
QUESTION 4

Which of the following substances has sulfur in its lowest oxidation state?

- (A) H_2S
- (B) SO_2
- (C) SO_4^{2-}
- (D) H_2SO_3

QUESTION 8

The graph below shows the pH changes during a titration process.



The graph represents the titration of a

- (A) weak acid with a strong base.
- (B) weak acid with a weak base.
- (C) strong acid with a strong base.
- (D) strong acid with a weak base.

QUESTION 9

Which of the following statements about the homologous series of aldehydes is correct?

- (A) The series consists of compounds of hydrogen and carbon only.
- (B) The formulas of successive compounds in the series differ by a $-\text{CH}_3$ group.
- (C) The series will have a trend of increasing boiling point with increasing size.
- (D) Isomers of members of the series may be described as primary, secondary or tertiary.

QUESTION 10

At 25°C , an aqueous solution will be acidic if

- (A) $[\text{H}_3\text{O}^+] > [\text{H}_2\text{O}]$
- (B) $[\text{OH}^-] = [\text{H}_3\text{O}^+]$
- (C) $[\text{H}_3\text{O}^+] > 10^{-7} \text{ M}$
- (D) $[\text{OH}^-] > 10^{-7} \text{ M}$

QUESTION 11

Which of the following amino acids can be classified as both diprotic and amphoteric?

- (A) alanine
- (B) aspartic acid
- (C) arginine
- (D) asparagine

QUESTION 12

Metal X was added to an aqueous solution, $\text{Q}(\text{NO}_3)_2$. A deposit formed on the surface of metal X.

Metal P was added to an aqueous solution, $\text{X}(\text{NO}_3)_2$. A deposit formed on the surface of metal P.

Deduce the order of reducing agent strengths for the three metals Q, P and X.

- (A) $\text{P} > \text{X} > \text{Q}$
- (B) $\text{X} > \text{P} > \text{Q}$
- (C) $\text{Q} > \text{X} > \text{P}$
- (D) $\text{Q} > \text{P} > \text{X}$

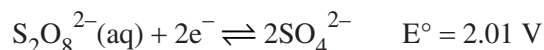
QUESTION 13

Which two compounds have the same molecular formula but different structural formulas?

- (A) butan-1-amine and butanenitrile
- (B) hexan-2-ol and pentanoic acid
- (C) 2-chlorobutane and 2-chloro-2-methylpropane
- (D) propane and propene

QUESTION 14

The following information is provided in addition to the information in the Formula and data booklet.



During the electrolysis of a dilute aqueous solution of copper(II) sulfate, sulfate ions are attracted to

- (A) the negative electrode, where they are reduced.
- (B) the positive electrode, where they are reduced.
- (C) the positive electrode, where they are oxidised.
- (D) the positive electrode, where they are neither oxidised nor reduced.

QUESTION 15

A primary alcohol, a secondary alcohol and a tertiary alcohol all have the same number of carbon, hydrogen and oxygen atoms per molecule.

The secondary alcohol

- (A) has a different molecular formula to the tertiary alcohol.
- (B) has a different empirical formula to the tertiary formula.
- (C) can be oxidised partially to give an aldehyde.
- (D) has two C atoms and an –OH group bonded to one carbon atom.

QUESTION 16

Which of the following statements applies to a galvanic cell but **not** an electrolytic cell?

- (A) The cathode is negatively charged.
- (B) Oxidation occurs at the anode.
- (C) Cations move towards the positive electrode.
- (D) The cell reaction converts electrical energy to chemical energy.

QUESTION 17

In two separate experiments, molten potassium chloride and 1.0 M potassium chloride solution were electrolysed using inert electrodes.

Potassium would be produced in

- (A) only one cell at the anode.
- (B) only one cell at the cathode.
- (C) both cells at the negative electrode.
- (D) neither cell.

QUESTION 18

A 0.200 M solution of a weak, monoprotic acid is found to have a pH of 2.92.

Deduce the K_a value for the acid.

- (A) 1.45×10^{-6}
- (B) 7.23×10^{-6}
- (C) 1.20×10^{-3}
- (D) 6.01×10^{-3}

QUESTION 19

Which of the following properties are generally characteristic of hydrocarbons with up to six carbon atoms per molecule?

- (A) low melting point and soluble in non-polar solvents
- (B) high melting point and soluble in non-polar solvents
- (C) low melting point and insoluble in non-polar solvents
- (D) high melting point and insoluble in non-polar solvents

QUESTION 20

Butan-2-one, $\text{CH}_3\text{COCH}_2\text{CH}_3$, is synthesised by the oxidation of butan-2-ol, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$.

Which of the following would be a suitable reagent for this conversion?

- (A) chlorine gas in the presence of ultraviolet light
- (B) acidified potassium dichromate solution
- (C) methanol in the presence of concentrated sulfuric acid
- (D) bromine water

QUESTION 21

Cellulose provides strength to cell walls.

Cellulose is able to form strong cell walls because

- (A) there is extensive branching in cellulose polysaccharide chains.
- (B) extensive covalent cross-links bond the cellulose polysaccharide chains to each other.
- (C) extensive hydrogen bonding occurs between cellulose polysaccharide chains.
- (D) the glycosidic links between the glucose monomers are stronger for cellulose than for other polysaccharides.

QUESTION 22

Identify the types of interactions that would be expected to occur between the side chains of glutamine and serine in the tertiary structure of a protein.

- (A) dispersion forces only
- (B) dipole–dipole forces only
- (C) hydrogen bonding only
- (D) hydrogen bonding and dispersion forces

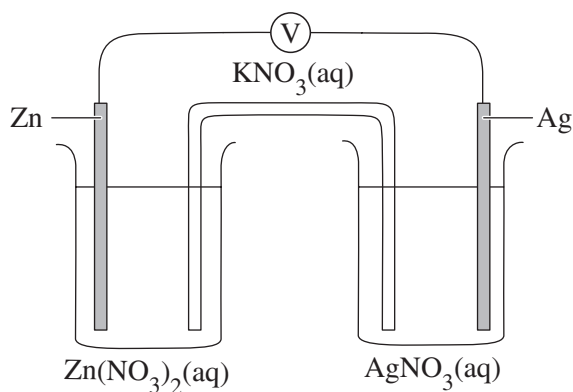
QUESTION 23

Apply your understanding of standard electrode potentials to identify a reagent that is capable of reducing Br_2 to Br^- but not capable of reducing I_2 to I^- under standard conditions.

- (A) Cl^-
- (B) Fe^{2+}
- (C) Ag^+
- (D) Cu

QUESTION 24

An electrochemical cell is constructed as shown below.

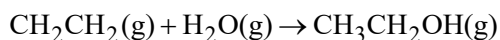


Which of the following statements about this electrochemical cell is correct?

- (A) Reduction occurs spontaneously at the positively charged Ag electrode.
- (B) Reduction occurs spontaneously at the negatively charged Ag electrode.
- (C) Reduction occurs non-spontaneously at the positively charged Ag electrode.
- (D) Reduction occurs non-spontaneously at the negatively charged Ag electrode.

QUESTION 25

A reaction involving organic compounds occurs according to the following equation.



Identify the term that could be used to describe this type of reaction.

- (A) substitution
- (B) redox
- (C) addition
- (D) condensation